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Analysis of A Deep Learning Algorithm for Fracture Detection in X-Ray Images

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ABSTRACT

Identifying bone fractures in X-ray images is a complex task that requires special expertise from radiologists and can be time-consuming in clinical workflows. Deep learning offers a significant automated diagnostic solution to improve accuracy and efficiency. This study aims to analyze the performance of three Convolutional Neural Network (CNN) architectures namely, ResNet50, DenseNet169, and EfficientNet-B3 and specifically compare the performance of models trained using augmented data with that of models trained without augmentation. The research method utilizes a local dataset, which is divided equally between the fractured and non-fractured classes. Preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) were applied, and the models were evaluated on a separate test set (hold-out test set). Model evaluation was conducted using accuracy, precision, recall, F1-score, ROC-AUC metrics, as well as analysis through confusion matrix, classification report, sensitivity, specificity, and calibration curve to assess overall performance. The experimental results show that the application of data augmentation consistently improves the accuracy and robustness of all three models. In the augmentation scenario, EfficientNet-B3 showed the best performance, achieving an accuracy of 93.33%. This study concludes that the combination of the EfficientNet-B3 architecture with the data augmentation strategy is the most optimal and recommended approach for developing a reliable automatic detection system on local X-ray image datasets.

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1. INTRODUCTION

A fracture is a medical condition in which the continuity of a bone is broken or cracked due to injury or accident, requiring early detection and timely treatment [1]. ne of the primary methods for detecting fractures is X-ray imaging, a medical imaging technique that utilizes X-ray radiation to reveal bone structures in detail [2]. Radiographic imaging (X-ray) examination is a procedure often relied upon for accurate assessment in diagnosing bone injuries, particularly to identify fractures (broken bones) [1].

X-ray image processing has undergone rapid development thanks to advances in digital technology [3]. This technology has become a crucial element in the medical world, as it enables healthcare professionals to diagnose various diseases accurately [3]. However, the process of manually analyzing X-ray images by radiologists often takes a long time and is highly dependent on the skill level of each radiologist [1], [2].

Advances in deep learning technology offer a more efficient and accurate alternative for automatically diagnosing bone fractures in X-ray images [4]. Additionally, deep learning can help overcome human limitations, such as potential fatigue and inconsistency resulting from varying levels of expertise in medical image interpretation [3]. The application of deep learning algorithms has the potential to help medical personnel speed up diagnosis, reduce errors, and improve decision-making accuracy [5].

The application of deep learning still faces several challenges, including the need for large amounts of training data for model development [5], [6]. This requirement is often constrained by issues of patient privacy and confidentiality [4]. Most of the available datasets originate from abroad [7], while the availability of local datasets in Indonesia is still limited. The use of foreign datasets is not necessarily optimal because the training data may not accurately represent the characteristics of the local population, which can lead to bias in data collection and result in incorrect conclusions [8]. Infrastructure limitations are also an obstacle to the application of deep learning [3], [4]. Therefore, the development of local datasets is essential to support model accuracy in the context of Indonesian data.

The Convolutional Neural Network (CNN) algorithm is a deep learning algorithm widely used in X-ray image analysis, as it can automatically extract features and classify bone fractures with high accuracy [9]. This study focuses on the analysis and evaluation of several algorithms developed from the CNN architecture, specifically Residual Neural Network (ResNet), Densely Connected Convolutional Networks (DenseNet), and EfficientNet, for detecting bone fracture cases in X-ray images.

The main issues to be examined in this study can be formulated as follows: (1) What are the steps for collecting and labeling a dataset of X-ray images of bone fractures? (2) What are the results of evaluating various deep learning architectures in detecting fractures, and which model has proven to be the most optimal for application to patient data in Indonesia? (3) How does the data augmentation technique affect the performance of deep learning models in detecting fractures?

Deep learning-based research often utilizes datasets from abroad [7], but the characteristics of foreign data may not be entirely suitable for Indonesian conditions [8]. Additionally, previous studies have focused primarily on a single algorithm without comprehensively comparing its performance with that of other algorithms. The novelty of this study lies in the use of a dataset obtained directly from hospitals in Indonesia, which accurately represents real conditions in local clinical practice. This study also analyzes and compares three CNN algorithms namely, ResNet, DenseNet, and EfficientNet to detect bone fractures in X-ray images.

2. RESEARCH METHOD

The research methodology consists of dataset collection, data preprocessing, data augmentation, model development and training, and model evaluation, as illustrated in Figure 1.

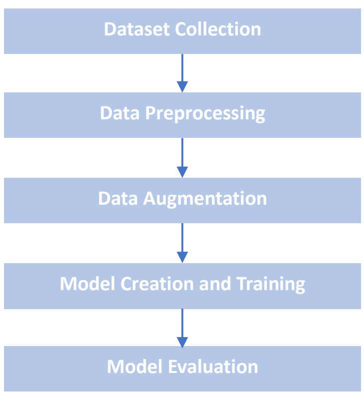

Figure 1. Research methodology.

2.1. Dataset Collection

This study uses a dataset collected from Karima Utama Hospital. The use of data from direct clinical sources aims to ensure that the developed model accurately represents real-world conditions in the field. All images collected were then converted to JPG format [10]. In order to maintain data privacy, all patients' personal identities were removed (anonymization). The entire data collection and anonymization process received ethical approval from the Health Research Ethics Committee, Faculty of Health Sciences, Universitas Muhammadiyah Surakarta (KEPK FIK UMS).

To ensure the accuracy and consistency of the labels on each image, the entire dataset was annotated manually. The annotators received training and direct guidance from a radiology specialist. This training aimed to standardize perceptions and establish a basic understanding of the visual characteristics of fractures, as well as how to interpret the clinical diagnoses associated with these images. The annotation process was carried out by referring to the patient's diagnosis recorded in the official medical report in the hospital medical record system. Each image was then processed and labeled according to the diagnosis results. In practice, challenges arise in some ambiguous cases, where the visual appearance of the image does not fully correspond to the diagnosis report. To ensure the accuracy of the labels, these cases are reviewed with an experienced radiologist who has in-depth expertise in musculoskeletal image interpretation, allowing for the consistent and accurate determination of the final labels. Through this process, the data was classified into two categories: fractured and non-fractured [11].

After annotation, the dataset used in this study consisted of a total of 1000 X-ray images with a balanced class distribution, specifically 500 images identified as fractured and 500 images identified as non-fractured. The data composition consists of 80% images from the extremities (hands, arms, shoulders, and feet) and 20% from other areas, such as the pelvis, spine, and collarbone. This composition directly reflects the frequency of cases handled at the medical institutions where the data was collected, where fractures of the extremities are more common. The dataset is then divided into two separate sets. A total of 700 images (70%), comprising 350 fractured images and 350 non-fractured images, formed the training pool (training dataset) used for the entire model training and validation process. The remaining 300 images (30% of the total), which also have a balanced distribution (150 fractured and 150 non-fractured), are allocated as a hold-out test set (a separate test dataset). This dataset is completely isolated and is only used in the final evaluation stage to measure the model's generalization performance [12]. Details of the dataset division are presented in Table 1.

Table 1. Distribution of the dataset.

Data Set	fractured images	non-fractured images	Number of Images
Training Pool	350	350	700
Test Set	150	150	300
Total	**500**	**500**	**1000**

2.2. Data Preprocessing

The data preprocessing stage is applied to address the challenges of image quality variation and normalize the data for optimal model training [4]. The first step is to enhance image contrast using Contrast Limited Adaptive Histogram Equalization (CLAHE). CLAHE is a highly effective technique for enhancing medical image quality by improving contrast and detail [13]. This technique aims to clarify important details in bone structures, such as fractures or fine cracks, which may be less visible in images with low contrast quality. Figure 2 shows the difference between the original image and the image that has undergone the CLAHE process.

After the visual quality is improved, the image then undergoes a series of standard transformations for the pre-trained ImageNet model. This process includes resizing the shortest side of the image to 256 pixels, cropping the central area to a final resolution of 224x224 pixels, converting the image to PyTorch tensor format, and normalizing the tensor using the mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] from the ImageNet dataset. To anticipate potential technical constraints, the system is also configured to process truncated images that may arise from the conversion process. After the visual quality is improved, the image then undergoes a series of standard transformations for the pre-trained ImageNet model. This process includes resizing the shortest side of the image to 256 pixels, cropping the central area to a final resolution of 224x224 pixels, converting the image to PyTorch tensor format, and normalizing the tensor using the mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] from the ImageNet dataset. To anticipate potential technical constraints, the system is also configured to process truncated images that may arise from the conversion process.

Figure 2. Image contrast enhancement using CLAHE.

2.3. Data Augmentation

The number of datasets collected was 1,000 images, which is relatively small for deep learning tasks. Therefore, an augmentation process was carried out to expand data variation and improve model generalization capabilities. Augmentation techniques are applied to overcome data limitations and reduce the risk of overfitting [13]. Augmentation is only performed on the training pool, while the hold-out test set remains unchanged to maintain the validity of the model evaluation. The techniques used included RandomHorizontalFlip with a probability of 50% (p=0.5) to make the model independent of orientation, RandomRotation up to 15 degrees to simulate minor variations in patient position, and RandomAffine, which includes shifts (translations) of up to 10% and scaling (zoom) between 90% and 110% to improve the model's robustness to object position and size. Through this technique, the number of training pool data increased from 700 original images to 5,000 augmented images (2,500 fractured and 2,500 non-fractured). A comparison of the composition of the training pool before and after this process is presented in Table 2. Thus, this study can compare the performance of models trained using data without augmentation with models trained using augmented data.

Table 2. Comparison of training pool size before and after augmentation.

Condition	fractured images	non-fractured images	Number of images
Before augmentation	350	350	700
After augmentation	2.500	2.500	5.000

2.4. Model Creation and Training

This stage aims to build, configure, and train deep learning models based on three main architectures, namely ResNet, DenseNet, and EfficientNet. Each architecture was selected because it has unique characteristics that are relevant to medical image classification, particularly in detecting bone fractures in X-ray images. The model formation process involves determining core parameters such as the optimizer, learning rate value, number of epochs, batch size, weight decay, and dropout rate, where the selection of these parameters is based on a literature review and further optimized through experiments to obtain the highest architecture effectiveness in the medical domain [10], [11], [14]. After the model was built, the training process was carried out using a cross-validation strategy to ensure the robustness and reliability of the model.

2.4.1. ResNet50

ResNet is a CNN architecture that utilizes residual connections to overcome the vanishing gradient problem, allowing for deep network training without compromising accuracy [15], [16]. This architecture uses shortcut connections that apply identity mapping to explicitly learn residual mapping, an approach that has been proven to facilitate optimization and accelerate convergence [17]. This study uses ResNet-50 because its architecture offers an excellent balance between significant depth increase for accurate performance and computational efficiency, achieved through the use of a proven, reliable 'bottleneck' design for medical image classification tasks [17].

The training configuration of this model includes the AdamW optimizer, a learning rate of 1×10^{-3}, a batch size of 16, a weight decay of 5×10^{-4}, a dropout rate of 0.5, and a total of 10 epochs, as shown in Table 3. The selection of these hyperparameters follows standard practices in medical image classification research using the ResNet architecture. It has been refined through preliminary experiments to ensure stable training and optimal model convergence. Similar configurations have also been adopted in previous studies employing ResNet for medical image classification, where the use of the AdamW optimizer or its variants, a learning rate of approximately 1×10^{-3}, a batch size of 16, and a dropout rate of 0.5 have been shown to enhance training stability and improve the model's generalization capability across complex medical image variations [18]–[20].

Table 3. Parameter configuration of the ResNet50 model.

Parameter	Value used
Optimizer	AdamW
Learning Rate	1×10^{-3}
Epoch	10
Batch Size	16
Weight Decay	5×10^{-4}
Dropout Rate	0.5

2.4.1. DenseNet169

DenseNet uses the concept of dense connectivity, where each layer receives input from all previous layers [21]. DenseNet has proven to be effective in medical image classification because its architecture creates a strong and stable flow of information through increased feature propagation efficiency and maximum feature reuse [22]. DenseNet-169 was chosen due to its efficiency advantages over other variants [23]. Table 4 presents the parameter configuration used in the DenseNet-169 algorithm, consisting of the AdamW optimizer, a learning rate of 1×10^{-3} , a batch size of 16, a weight decay of 5×10^{-4}, a dropout rate of 0.5, and a total of 10 epochs. The selection of these parameters is based on standard practices adopted in medical image classification studies employing the DenseNet architecture. Previous works have shown that the combination of adaptive optimizers and regularization techniques, such as dropout, effectively stabilizes training and enhances model generalization in medical imaging tasks [24], [25]. The dropout rate of 0.5 is applied to randomly deactivate half of the neurons during training, which helps reduce co-adaptation between features and prevents overfitting on limited datasets [25]. Experimental findings in related studies demonstrate that such configurations lead to stable convergence and balanced performance between training efficiency and model generalization [24].

Table 4. Parameter configuration of the DenseNet169 model.

Parameter	Value used
Optimizer	AdamW
Learning Rate	1×10^{-3}
Epoch	10
Batch Size	16
Weight Decay	5×10^{-4}
Dropout Rate	0.5

2.4.1. EfficientNet-B3

EfficientNet is a CNN architecture that employs the compound scaling principle, a technique that simultaneously scales the depth, width, and resolution of the network, aiming to achieve optimal parameter efficiency without compromising accuracy [26]. This capability enables the architecture to produce comprehensive feature representations that summarize both high-level abstractions and the nuances of detail necessary to distinguish between cracked and non-cracked images [27]. This study uses EfficientNet-B3 because this variant is an advanced deep convolutional neural network architecture recognized for its ability to achieve high accuracy while optimizing computational efficiency [27]. This capability is rooted in the hierarchical feature learning process, which enables the model to automatically extract and recognize complex patterns from image data [26].

The training configuration includes the AdamW optimizer, a learning rate of 1×10^{-3}, a batch size of 16, 10 epochs, a weight decay of 1×10^{-3}, and a dropout rate of 0.45, as shown in Table 5. The learning rate of 1×10^{-3} has been shown to provide stable convergence and high accuracy in medical image classification tasks [28]. The AdamW optimizer is employed to decouple the weight decay from the gradient-based parameter updates, leading to a more regularized and efficient learning process. A weight decay of 1×10^{-3} is applied as a form of regularization to control model complexity. At the same time, a dropout rate of 0.45 is selected to allow the model to learn complex feature patterns without compromising training stability.

Table 5. Parameter configuration of the EfficientNet-B3 model.

Parameter	Value used
Optimizer	AdamW
Learning Rate	1×10^{-3}
Epoch	10
Batch Size	16
Weight Decay	1×10^{-3}
Dropout Rate	0.45

2.4.4. Model Training

The model training process is designed to test each data sample evenly and prevent bias errors in its performance evaluation [29]. For this purpose, the stratified K-fold cross-validation methodology with k = 5 is applied to the Training Dataset. Cross-validation here serves as a robust training strategy to produce five reliable candidate models, rather than a final evaluation method. The use of "Stratified" ensures that the proportion of fracture and non-fracture classes (50:50) is maintained in each fold, which is crucial for preventing training bias and ensuring that each fold accurately represents the overall data distribution. In each of the five iterations, the model is trained on 4,000 images and validated on 1,000 images.

This study applies optimization and regularization techniques to improve the quality and efficiency of the training process. The Early Stopping technique is used to automatically terminate training when the model's performance on the validation dataset begins to decline, serving as a form of regularization to prevent overfitting [23]. Model Checkpointing is the process of saving a "snapshot" of the model weights when its performance is at its best. Specifically, every time the validation loss at the end of an epoch is lower than the best loss previously recorded, the model weights at that time are saved. As a result, the model used for the final evaluation is not the model from the last epoch, but the model that shows the best performance [30].

2.5. Model Evaluation

The evaluation stage aims to provide an objective and reliable assessment of model performance [31]. To ensure robust model development, this study conducted a final evaluation on an independent test dataset, a crucial step to prevent data leakage from the augmentation results into the validation set and maintain the integrity of the evaluation [5].

The evaluation protocol implemented is as follows: The five best models, which have been saved through the model checkpointing mechanism from each of the five training folds, are taken as the final candidate models. Each of these candidate models is then tested on a hold-out test set consisting of 300 images (150 fractured and 150 non-fractured). This test data was never used during the training or validation process, so the evaluation truly reflects the model's generalization ability on new data [30], [31].

Each image in the test set generates a prediction score with a value range of 0 to 1, indicating the probability that the image belongs to the fractured or non-fractured category. After all scores are obtained, the model performance is calculated using quantitative evaluation metrics, namely accuracy, precision, recall, and F1-score [5]. Among these metrics, the F1-score is emphasized due to its ability to balance precision and recall, which is particularly important for medical classification tasks where accurate predictions for both positive and negative cases carry different significance weights [5].

The calculation formulas for each evaluation metric are described as follows :

Accuracy is calculated to determine the percentage of correct model predictions against all test data, as shown in Equation (1).

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

Precision is used to measure the model's accuracy in predicting positive data (fracture), as shown in Equation (2).

$$Precision = \frac{TP+TN}{TP+FP} \quad (2)$$

Recall is used to measure the model's sensitivity in identifying all positive data (fracture), as shown in Equation (3).

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

F1-score is used to obtain the harmonic mean between precision and recall, as shown in Equation (4).

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision+Recall} \quad (4)$$

Where:

- TP (True Positive): Positive data that is correctly predicted.
- TN (True Negative): Negative data that is correctly predicted.
- FP (False Positive): Negative data that is incorrectly predicted as positive.
- FN (False Negative): Positive data that is incorrectly predicted as negative.

As a step in in-depth analysis, a classification report is also included, which provides details on the precision, recall, and F1-score values for each class. To visualize the distribution of correct and incorrect predictions in more detail, a confusion matrix is used to evaluate the model's performance by summarizing the prediction results against the actual data labels, which helps analyze the strengths and weaknesses of the classification model [12]. An evaluation was also conducted, including sensitivity and specificity to assess the balance of detection between classes, Receiver Operating Characteristic–Area Under Curve (ROC-AUC) to measure the discriminatory ability of the model, and a calibration curve to assess the suitability between the predicted probability and the actual results. Next, the model with the best performance from each architecture is selected for direct comparison. The best model is the one that achieves the highest accuracy value on the test data [14]. This approach was chosen to show the peak performance and optimal potential that can be achieved by each architecture within the established experimental framework. The results of the two dataset scenarios (without augmentation and with augmentation) are compared systematically. This comparison enables researchers to empirically evaluate whether data augmentation indeed yields a significant improvement in model generalization or has a minimal impact.

3. RESULTS AND DISCUSSION

The performance of the three CNN architectures was evaluated by comparing models trained in two different scenarios: one using training data without augmentation and the other using augmented training data. Table 6 summarizes the results of comparing the performance of each model in both scenarios.

Table 6. Model evaluation comparison with and without augmentation.

Model	Augmentation	Accuracy	Precision	Recall	F1-Score
ResNet50	No	88,33%	88,54%	88,33%	88,32%
	Yes	90,67%	90,70%	90,67%	90,67%
DenseNet169	No	81%	83,25%	81%	80,67%
	Yes	86,33%	87,55%	86,33%	86,22%
EfficientNet-B3	No	91,33%	91,36%	91,33%	91,33%
	Yes	93,33%	93,83%	93,33%	93,31%

3.1. ResNet50

Under conditions without data augmentation, ResNet50 produced a solid baseline performance with an accuracy of 88.33%, precision of 88.54%, recall of 88.33%, and an F1-score of 88.32%. These values were achieved thanks to the fundamental strength of the ResNet50 architecture. With the concept of residual learning through skip connections, this model effectively trains deep networks, making it easier to optimize and achieve accuracy improvements with greater depth [17]. This capability is particularly relevant in medical research, as it can be used to capture complex patterns in X-ray images of fractured. The slight difference where the precision value (88.54%) is higher than the recall (88.33%) indicates that the model tends to be more cautious; when uncertain, the model prefers to predict "non-fractured" to avoid false positives, even though this risks missing some cases of actual fractured (false negatives).

After applying the data augmentation technique, a significant performance improvement was observed across all metrics, as shown in Figure 3. The ResNet50 model trained with augmented data achieved an accuracy of 90.67%, precision of 90.70%, recall of 90.67%, and an F1-score of 90.67%. This performance improvement was achieved through data augmentation techniques that enriched the training data with variations that retained the labels, such as flipping and rotation. This approach is an effective solution to the problem of data limitations, especially in the medical imaging domain [32].

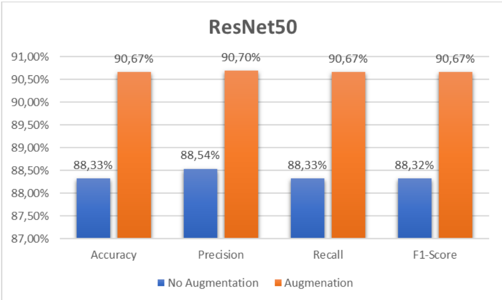

Figure 3. Experimental results of the ResNet50 model.

This process forces the model to learn to recognize fracture features that are more robust and invariant to position or orientation, thereby directly improving its generalization ability and reducing overfitting. As a result, the model not only becomes more accurate overall but also achieves a near-perfect balance between precision and recall. This shows that the model becomes more confident in identifying actual fracture cases (increased recall) without sacrificing its ability to avoid false positive diagnoses (maintaining high precision).

The classification report in Table 7 indicates that ResNet50 achieves an accuracy of 90.67% and exhibits a balanced distribution across both classes. The precision for the fractured class is 0.92, with a recall of 0.89. Conversely, the non-fractured class has a precision of 0.90 with a recall of 0.92. The identical F1-score values (0.91) for both classes indicate that the model does not have a significant bias towards either class.

Table 7. Classification report of the ResNet50 model.

Class	Precison	Recall	F1-Score	Support
fractured	0.92	0.89	0.91	150
non-fractured	0.90	0.92	0.91	150
Accuracy			0.91	300
Macro Avg	0.91	0.91	0.91	300
Weighted Avg	0.91	0.91	0.91	300

The distribution of model errors is clarified through the Confusion Matrix in Figure 4, which shows that the model makes two main types of errors in relatively equal numbers: 16 fractured cases are misclassified as non-fractured (false negatives), and 12 non-fractured cases are incorrectly predicted as fractured (false positives). This balance between false positives and false negatives indicates that the ResNet50 model, after augmentation, successfully achieved a good compromise between sensitivity and specificity without leaning excessively toward either type of error.

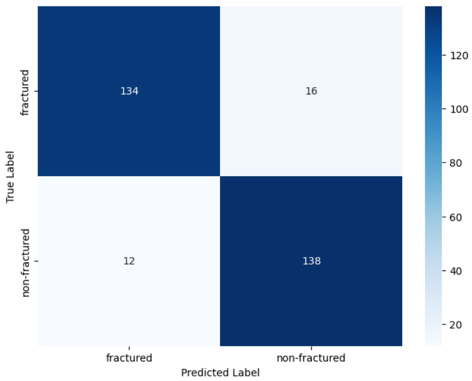

Figure 4. Confusion matrix of the ResNet50 model.

The balance of the model's performance in detecting fractures and accurately identifying non-fractures is shown in Figure 5. For the 'fractured' class, the model shows strong detection ability (sensitivity) of 0.89. Interestingly, its ability to avoid misclassifying non-fracture cases (specificity) is even higher, reaching 0.92. A similar pattern is seen in the 'non-fractured' class, where the sensitivity is 0.92 and the specificity is 0.89. The balance of these values highlights the model's reliable performance and lack of significant bias towards either class.

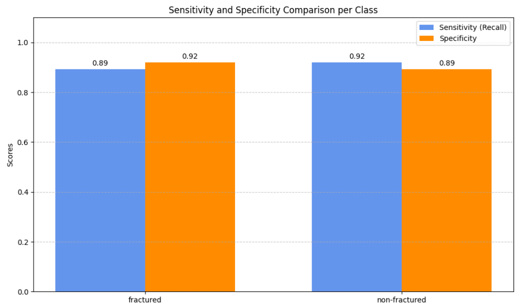

Figure 5. Sensitivity and specificity of the EfficientNet-B3 model.

The model's ability to clearly distinguish between fracture and non-fracture cases is firmly validated by the ROC curve in Figure 6. With an impressive AUC score of 0.96, this model has demonstrated high reliability. Visually, the curve's trajectory, soaring high to the upper left corner, far beyond the random line, illustrates its ability to maximize the actual positive rate while minimizing the false positive rate.

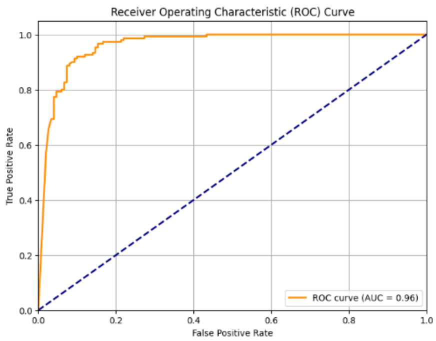

Figure 6. ROC curve of the ResNet50 model.

In addition to accuracy, it is also essential to know how much we can trust the confidence level of the model's predictions. The calibration diagram in Figure 7 evaluates the reliability of the confidence level of the model's predictions. The dotted diagonal line on this plot represents perfect calibration, where the probability of the prediction ideally equals the frequency of the actual result.

The plot results show that the model tends to be slightly overconfident when predicting the 'fractured' class (blue curve below the ideal line), but somewhat underconfident for the 'non-fractured' class (orange curve above the perfect line). Nevertheless, the trend of both curves following the diagonal line indicates that the model's confidence has a good correlation with the actual results.

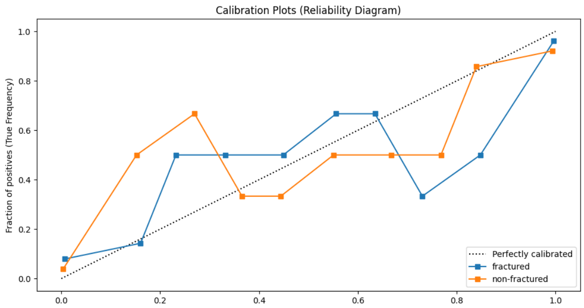

Figure 7. Calibration curve of the ResNet50 model.

The good correlation between model predictions and actual results, as well as its overall balanced performance, can be attributed to a key factor: the use of a base model whose weights have been pre-trained on the massive ImageNet dataset. By utilizing the mature "ResNet50 Base Model" framework, this architecture does not require training from scratch; instead, it only needs to be adjusted by adding a final processing layer to adapt it to the specific task of fracture recognition [22]. More fundamentally, the reliability of the ResNet architecture is rooted in its core design, which uses residual blocks to mitigate the vanishing gradient problem. This capability is crucial because it enables stable training of intense networks, allowing them to learn complex visual features that are essential for transfer learning tasks on medical images [15]. The strong performance of this approach is also quantitatively validated in another study, which demonstrates that the ResNet-50 model consistently achieves an average score of 0.93 for precision, recall, and F1-score metrics. This figure confirms its capability as a highly reliable architecture for medical image classification [33].

3.2. DenseNet169

DenseNet169 recorded the lowest performance among the three architectures tested. In the scenario without augmentation, this model achieved an accuracy of 81.00%, precision of 83.25%, recall of 81.00%, and an F1-score of 80.67%. This finding aligns with other comparative studies, which show that although DenseNet's performance is relatively stable, it still falls short of that of other architectures. This suboptimal performance can be attributed to a common challenge in deep learning, where a limited amount of training data can cause the model to struggle to generalize features effectively and be prone to overfitting [23]. This limitation is recognized as a factor that can limit performance, which can be improved by adding more data samples [21].

With the application of data augmentation, as shown in Figure 8, the model's performance improved significantly to an accuracy of 86.33%, a precision of 87.55%, a recall of 86.33%, and an F1-score of 86.22%. This increase in accuracy of more than 5% demonstrates that data augmentation is highly effective in enabling DenseNet169 to learn more generalizable feature representations. Augmentation successfully rebalanced the model's performance to a higher and more reliable level. However, the final results are still below those of other architectures, indicating that DenseNet169 may be less than optimal for the specific characteristics of this dataset, even after being enriched with synthetic data.

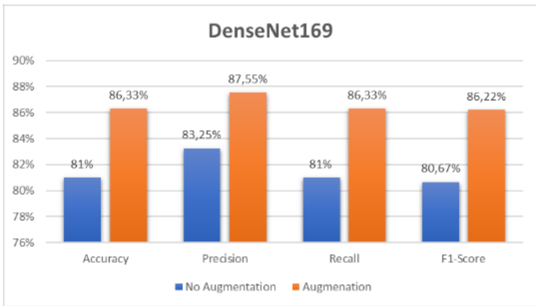

Figure 8. Experimental results of the DenseNet169 model.

Despite achieving 86% accuracy, analysis of the classification report in Table 8 shows a significant imbalance between precision and recall in both classes.

Table 8. Classification report of the DenseNet169 model.

Class	Precison	Recall	F1-Score	Support
fractured	0.94	0.77	0.85	150
non-fractured	0.81	0.95	0.87	150
Accuracy			0.86	300
Macro Avg	0.88	0.86	0.86	300
Weighted Avg	0.88	0.86	0.86	300

The fractured class shows high precision (0.94) but lower recall (0.77). This indicates that although the fractured predictions made by the model tend to be correct, the model still misses about 23% of actual fractured cases. This pattern is further confirmed by the Confusion Matrix in Figure 9, where the most dominant error is false negatives, with 34 fractured cases misclassified as non-fractured. Meanwhile, 7 non-fractured cases were incorrectly predicted as fractured.

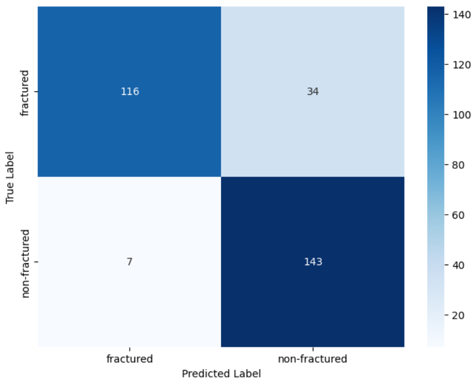

Figure 9. Confusion matrix of the DenseNet169 model.

As shown in Figure 10, a comparative analysis of sensitivity and specificity for each class reveals different performance profiles. For the 'fractured' class, this model exhibits a very high specificity of 0.95, indicating that it is highly reliable in correctly identifying non-fracture cases. However, the sensitivity (recall) for this class is 0.77, indicating its ability to detect most actual fracture cases. Conversely, for the 'non-fractured' class, the sensitivity is very high at 0.95, while the specificity is 0.77.

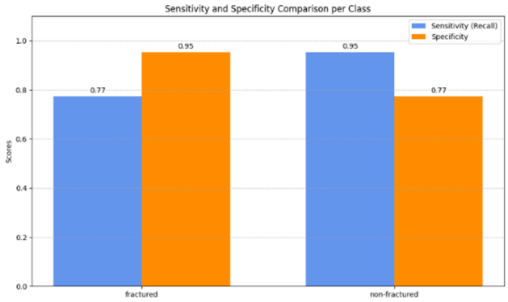

Figure 10. Sensitivity and specificity of the DenseNet169 model.

The ROC curve presented in Figure 11 confirms the model's excellent discriminatory ability. With an AUC value of 0.95, this curve shows that the model has a strong ability to distinguish between the 'fractured' and 'non-fractured' classes. The curve's position close to the upper left corner visually confirms a high classification success rate with a low false positive rate.

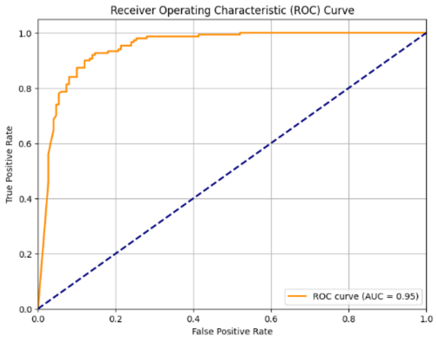

Figure 11. ROC curve of the DenseNet169 model.

Figure 12 presents a calibration curve that illustrates the reliability of the probabilities predicted by the model. For the 'fractured' class (blue), the curve is above the perfect calibration line, indicating that the model tends to be slightly underconfident when predicting fractures. Conversely, for the 'non-fractured' class (orange), the curve generally lies below the ideal line, indicating a tendency to be overconfident in its predictions. Although not perfectly calibrated, both curves follow a diagonal trend, indicating that the model's probability confidence is consistent with the actual results.

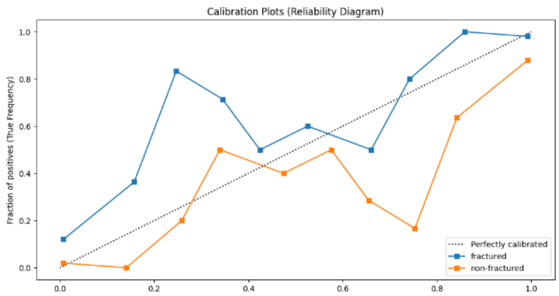

Figure 12. Calibration curve of the DenseNet169 model.

This calibration behavior, particularly the underconfidence tendency in the 'fractured' class, leads to a deeper analysis of how the DenseNet architecture affects prediction results. Architecturally, DenseNet is designed to be highly efficient by connecting each layer to all subsequent layers, which has been shown to improve parameter efficiency and gradient flow by promoting extensive feature reuse [21]. However, in this study, this characteristic does not provide maximum benefits. One possible reason is that the DenseNet architecture is highly dependent on the quality of features from the early layers. Suppose the features extracted in the early stages are not discriminative enough to detect subtle fractures. In that case, the extensive propagation of these less informative features can actually interfere with the learning process in the deeper layers.

3.3. EfficientNet-B3

EfficientNet-B3 showed the best results in both testing scenarios, proving its superiority as the most optimal architecture in this study. Without augmentation, this model achieved an accuracy of 91.33%, precision of 91.36%, recall of 91.33%, and an F1-score of 91.33%, surpassing the baseline performance of ResNet50 and DenseNet169. This high initial performance can be attributed to the EfficientNet architecture design, which is inherently efficient and capable of achieving high accuracy with fewer computational resources, making it highly effective even on limited datasets [26].

With the application of data augmentation, as shown in Figure 13, the performance of EfficientNet-B3 improved even more significantly, achieving an accuracy of 93.33%, a precision of 93.83%, a recall of 93.33%, and an F1-score of 93.31%. The application of data augmentation has proven to be an effective strategy for artificially enriching datasets, enabling models to utilize additional data variations to improve generalization and robustness, especially in domains with limited data[32] .

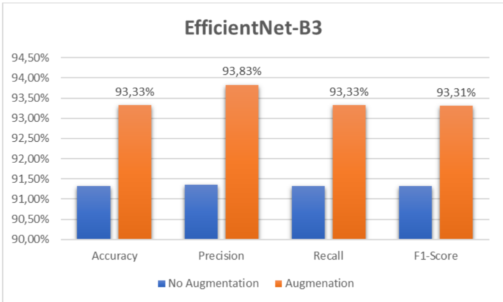

Figure 13. Experimental results of the EfficientNet-B3 model.

The very high precision value (93.83%) is crucial in a clinical context, as it demonstrates the model's effectiveness in minimizing false positives, thereby reducing the likelihood of patients receiving incorrect diagnoses. On the other hand, the equally strong recall (93.33%) confirms the model's ability to capture most actual bone fracture cases, minimizing the risk of missed cases. The remarkable balance between these two metrics, reflected in the high F1-score, makes it a highly reliable model for diagnostic applications.

The model's performance is further analyzed through the Classification Report presented in Table 9. The classification report shows very high precision for the fractured class (0.99), indicating that when the model predicts a fractured, the prediction is almost always correct. For the non-fractured class, it achieves near-perfect recall (0.99), indicating that the model is highly reliable in identifying all cases that are genuinely non-fractured.

Table 9. Classification report of the EfficientNet-B3 model.

Class	Precison	Recall	F1-Score	Support
fractured	0.99	0.88	0.93	150
non-fractured	0.89	0.99	0.94	150
Accuracy			0.93	300
Macro Avg	0.94	0.93	0.93	300
Weighted Avg	0.94	0.93	0.93	300

The Confusion Matrix reinforces this analysis in Figure 14. The matrix indicates that the model's primary error pattern is False Negative, where 18 fractured cases were incorrectly classified as not having a fractured. On the other hand, the false positive rate is very low, with only

2 non-fractured cases incorrectly identified as fractured. This finding confirms that the model tends to be very cautious in predicting fractured. This is very effective in minimizing false positive diagnoses, albeit at the risk of missing some cases that are actually positive.

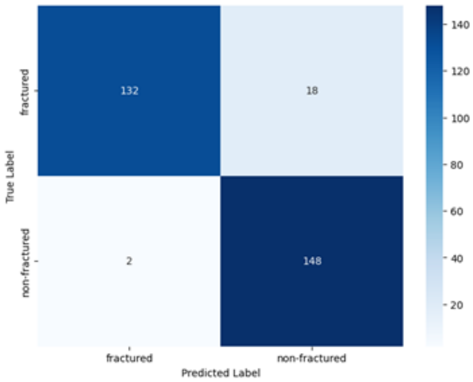

Figure 14. Confusion matrix of the EfficientNet-B3 model.

A more detailed analysis for each class shows that the model is highly reliable in correctly identifying non-fracture cases. Figure 15 details that for the 'fractured' class, the model achieved a Sensitivity (Recall) of 0.88 and a Specificity of 0.99. This very high specificity value confirms that the risk of false positives (misdiagnosing non-fractures as fractures) is very low.

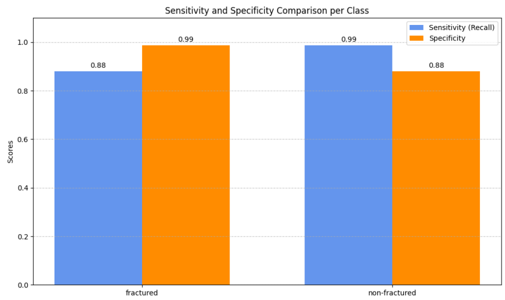

Figure 15. Sensitivity and specificity of the EfficientNet-B3 model.

The discrimination performance of the model was evaluated using an ROC curve, as shown in Figure 16. This figure shows an AUC (value of 0.98, which is very close to the perfect value of 1.0. This score indicates that the model has an exceptional ability to distinguish between fracture and non-fracture classes accurately.

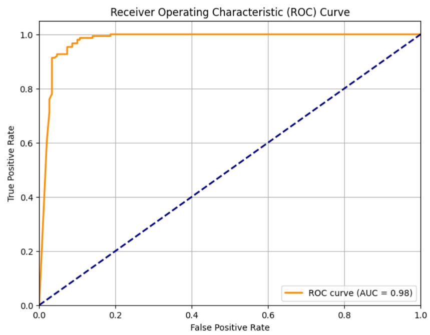

Figure 16. ROC curve of the EfficientNet-B3 model.

Calibration plots are presented to assess the reliability of the probabilities generated by the model, as shown in Figure 17. These plots compare the predicted probabilities with the actual frequencies. It can be seen that the curves for the 'fractured' and 'non-fractured' classes show some deviation from the perfect calibration line. This indicates that although the model is very accurate in distinguishing between classes, the confidence levels it means are not always consistent with the actual probabilities of the results.

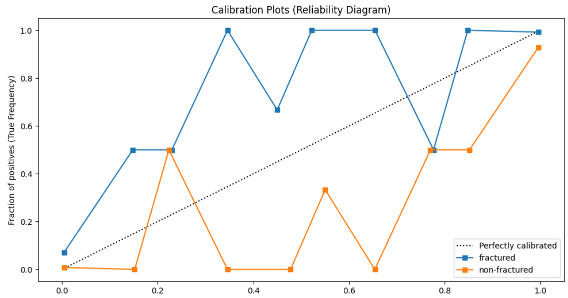

Figure 17. Calibration curve of the EfficientNet-B3 model.

Despite this imperfect probability calibration, the highly accurate model's discriminative performance is rooted in the fundamental strengths of its architecture. The fundamental advantage of EfficientNet lies in its compound scaling strategy. EfficientNet uses a compound scaling methodology that aligns the depth, width, and resolution of the network, making it highly adept at tasks involving image classification [27]. In addition, the effectiveness of this architecture is enhanced through transfer learning, where a model trained on a large image dataset is then fine-tuned on a specific radiography dataset to extract relevant features. This holistic approach enables the model to extract richer and more complex hierarchical features without drastically increasing the computational load. It is this capability that makes EfficientNet-B3 highly effective at learning strong feature representations, which are essential for subtle detection tasks such as identifying fractured lines in medical images [27].

3.4. Discussion

This study shows that the EfficientNet-B3 architecture consistently delivers the best performance compared to ResNet50 and DenseNet169 in terms of accuracy, precision, recall, and F1-score for bone fractured detection tasks. With an accuracy of 93.33%, EfficientNet-B3 demonstrates a significant advantage. These results are in line with the general principle in deep learning where newer architectures are often designed to achieve an optimal trade-off between accuracy and computational efficiency, an important consideration in real-world medical imaging applications [26], [27].

EfficientNet-B3 also exhibits a better balance between precision and recall, as evidenced by the highest F1-score value (93.31%). This balance is critical in a medical context because the model not only accurately identifies bone fractured cases (high recall) but also minimizes false positive diagnoses (high precision). The importance of the F1-score as a critical metric in evaluating medical classification models has been emphasized in various evaluation studies, which aim to provide a balanced performance overview [1], [12].

To gain a deeper understanding of the model's limitations, an analysis of misclassified cases was conducted, as shown in Figure 18. This analysis revealed that the errors made by the model were not random but followed a systematic pattern closely related to the anatomical complexity and subtlety of features in X-ray images. The model's errors were concentrated in areas of highly complex anatomy, where approximately half of the 20 misclassified images were of the hands and feet, while the rest were of complex joints, such as the elbows and shoulders. The leading cause of errors in these areas was the large number of small bones that were densely packed and close together, creating a visually dense image. This visual density obscures fracture lines, making them difficult to distinguish from normal boundaries between bones. As a result, the model struggles to isolate these abnormalities, a challenge that requires it to recognize intricate patterns in medical images [22].

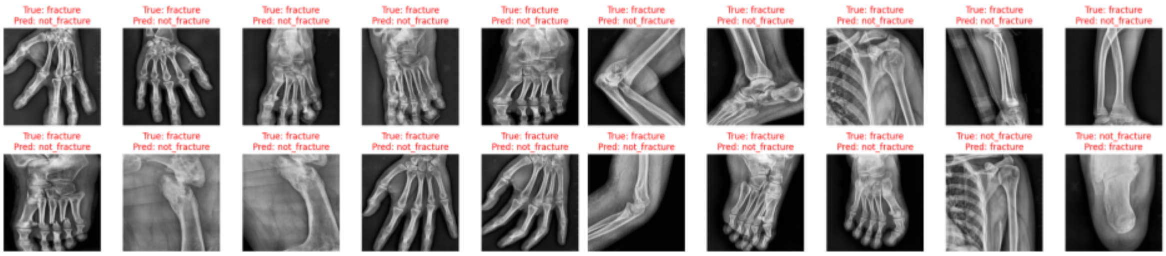

Figure 18. Misclassified samples of the EfficientNet-B3 model.

This challenge is even greater when dealing with features that resemble each other. In some cases, particularly in images of the wrist and elbow, classification errors are also influenced by the nature of the fracture itself, which resembles a fine crack or hairline fracture. The ability to detect faint fracture lines like this is a crucial benchmark for evaluating the capabilities of a deep learning model. On the other hand, this phenomenon of "copycat features" also causes false positives, where the model mistakenly identifies regular growth lines on the bone as fractured. This again highlights the challenge of distinguishing normal anatomical variations from pathological abnormalities, where the model must be able to distinguish subtle details to avoid errors [34]. Hus, the difficulty in model classification stems from specific challenges in interpreting very subtle fracture features. One study concluded that the characteristics of small fracture lines within complex anatomical structures can be the leading cause of classification difficulties [34], as such subtle features are often difficult to recognize even with traditional diagnostic techniques [27].

These findings are consistent with previous reports highlighting limitations in specific architectures. Several studies have reported that architectures such as ResNet and DenseNet can show unsatisfactory performance, especially in complex medical image classification scenarios [2], [34]. This contrasts with more modern architectures, such as EfficientNet, which, in comparative studies, have proven to be superior. An EfficientNet-B3-based model achieved 93.5% accuracy, surpassing the ResNet-50-based model at 90.3% [27]. Furthermore, other studies also confirm that the deep learning approach can generalize well across various medical imaging modalities [4], thereby strengthening the relevance of these research results in a clinical context. This comparison shows that although ResNet and DenseNet remain relevant, the efficiency and accuracy of EfficientNet make it superior for implementation.

Overall, the results of this study reinforce empirical evidence that the EfficientNet-B3 architecture has great potential in medical image-based diagnosis, not only for fracture detection, but also for other domains such as pneumonia, tumors, and abnormalities in CT scan images. This finding aligns with general observations that confirm the potential of deep learning in medical imaging transformation [35]. With these achievements, this research is expected to provide a strong foundation for the development of efficient and accurate deep learning-based clinical decision support systems.

4. CONCLUSION

This study classified X-ray images to detect bone fractured by evaluating the performance of deep learning architectures: ResNet50, DenseNet169, and EfficientNet-B3. The dataset used was sourced from hospitals in Indonesia and consisted of 1,000 manually annotated images, which were expanded to 5,000 images through augmentation. The results of the experiment show that all architectures are capable of detecting bone fractured with an accuracy above 80%. EfficientNet-B3 demonstrates the best performance, achieving an accuracy of 93.33%, precision of 93.83%, recall of 93.33%, and F1-score of 93.31%. The application of data augmentation has been proven to improve the robustness of the model and its ability to handle image variations, especially in limited datasets.

Although the results are promising, this study has two main limitations. First, the use of a dataset from a single institution has the potential to limit data variability. Second, the model only classifies images as a whole without localizing specific fracture areas. Therefore, further research is recommended to utilize multi-institutional datasets to enhance generalization and develop object detection functionality that localizes fractures with precision, thereby supporting more comprehensive clinical practices.

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